SENTINEL AI

Phase 1 Completion Report

Deterministic control path, live stream and Operations Control Tower — verified against the v1.2 Master, Technical and Toolchain & Compliance documents



Competition	AI Builder Cup 2026 · BFSI — Intelligent Risk, Fraud & Financial Experiences
Report date	13 October 2026 (freeze 15 October, submission 18 October)
Repository	github.com/Mohammed-Ilyas-AI/Sential-AI · branch `main` · last commit `26ebe5d` plus the Phase 1 close-out changes in this report
Evidence	Code inspection, 219 automated tests, 25 integration tests on MongoDB Atlas, and screenshots of the running application with live Atlas data

1. Executive summary

Phase 1 delivers the part of SENTINEL AI that the v1.2 documents call non-negotiable: **rules decide**. A transfer reaches `POST /v1/decision/transactions`, is scored by seven deterministic rules, passes the SG-RD-50 rapid-drain gate, is persisted with a versioned and reproducible decision, and appears on the Operations Control Tower within the same second through MongoDB change streams and a WebSocket. The canonical demo scenario, DEMO_TAKEOVER_01, produces exactly the documented numbers — scores 37, 49, 75 and 95, a 65.4% drain of a S$52,000 balance and a 24-hour HOLD — and an automated test proves it on every run.

3 of 8	**2 of 8**	**244**	**0**
MVP modules done (M§7)	MVP modules partial	tests passing (219 + 25 on Atlas)	secrets in the repository

> **What is not built yet, and why it matters.** The intelligence path — Gemini on Vertex AI, the investigation workspace, the entity-graph screen and the case report — is designed and wired as placeholders but does not run. The Toolchain document lists *meaningful use of Google AI* and *deployment on Google Cloud* as entry requirements (C§2), and Technical Merit & GenAI carries 40% of the score (M§4). Phase 2 must start there. Section 8 sets out the order.

2. How this report was verified

- **Documents.** The three v1.2 PDFs in `docs/` were re-read in full; every requirement below cites its section (M = Master, T = Technical, C = Toolchain & Compliance).
- **Code.** Each claim was checked in the source, not from memory — for example the absence of a kill switch and of the hold cascade was confirmed by searching the code.
- **Running system.** Screenshots are of the real application: Angular UI, `sentinel-core` and `sentinel-stream` running locally against the MongoDB Atlas cluster `sentinel_ai` (Google Cloud, asia-southeast1), after the simulator played DEMO_TAKEOVER_01.
- **Tests.** `npm run verify` (typecheck, lint with architecture-boundary rules, all tests, all builds) and `npm run test:integration` against Atlas — both green on Node 24.18.0.

3. MVP modules — Master Document §7

Module	Status	What exists, with evidence	Gap
1 Transaction simulator	Done	Canonical `TransactionEvent` with tokenised IDs, `eventId`, per-account sequence and idempotency key; seeded generators (baseline, takeover, mule network) and DEMO_TAKEOVER_01; Pub/Sub publisher with ordering key `sourceAccountToken`, plus direct push for local runs. `npm run simulate`. 6 + 13 tests.	Noise generator not implemented. One-button scenario launcher in the UI is a stretch item (M§7).

2 Real-time risk engine	Done	Seven pure rules (rapid drain, new device, new payee, velocity, entity link, behavioural deviation, cooling-off) with curves ported from the HTML behavioural reference (ADR-006). `POST /v1/decision/transactions` answers ALLOW / HOLD / REJECT with ruleset and policy versions, trace ID and expiry. 62 rule tests.	—
3 Rapid-drain detection	Done	S$50,000 protected balance, strictly more than 50% of the balance at window start, 24-hour rolling window that counts each payment once; exclusions for standing instructions, GIRO and designated billing organisations. Policy values are configuration (SG-RD-1.0).	Hold of *subsequent* outflows after the gate fires (M§3, T§4) is not implemented.
4 Intervention workflow	Partial	Automatic HOLD with decision trace; append-only audit trail with actor, reason codes and assessment snapshot; 24-hour auto-release scheduler (×3600 in demo time, labelled in the UI); APIs for customer-verified release, customer cancellation and analyst fraud confirmation with analyst identity; mock customer notification.	No analyst action buttons in the UI; **Contact customer** and **Kill switch** not built.
5 Investigation workspace	Not started	Route, page and component shells; evidence-bundle compiler port defined.	Evidence bundle assembly and display.
6 Gemini explanation	Not started	Model router (model IDs from configuration), versioned system instruction, response-schema and source-field validator shells; lint rule keeps AI out of the control path.	No Vertex AI call yet; no schema-validated case summary.
7 Entity graph	Partial	Backend: `POST /v1/network/observations`, `entities` and `entity_edges` with edge evidence; two-hop flagged-link query feeding the ENTITY_LINK rule (weakest link on the path).	Interactive graph screen, edge provenance panel, trace funds and the observed boundary.
8 Case report	Not started	Page and component shells.	One-click report, edit, sign-off, export.

4. Surfaces and demo scenes — Master Document §5 and §6

Surface (M§5)	Status	Where, and notes
5.1 Live Operations Command Center	Done	Control Tower. Counters follow the approved visual design (live transactions, high-risk alerts, held transactions, monitored banks) rather than the four listed in M§5.1; value protected and transactions per second can be restored on request.
5.2 Account risk panel	Done	Alerts & Cases: balance at window start, 24-hour outflow, drain against the 50% line, score and band, the five binary signals.
5.3 Intervention	Partial	Decision trace (rule, evidence, source, versions) and intervention state are shown; analyst actions are not.
5.4 Investigation workspace	Not started	Page shell with an honest availability note.
5.5 Network analysis	Not started	Page shell; scoring already uses the graph.

5.6 Case report	Not started	Page shell.

#	Demo scene (M§6)	Ready	Notes
1	Normal operations	Yes	Baseline transfers stream in LOW and allowed.
2	Attack begins	Yes	New device and new payee events; transfers accelerate.
3	Detection, 21 → 95	Yes	Scores 37, 49, 75, 95 on the stream and the alert; the 21 and 37 posture scores are recorded but not shown as their own rows.
4	Threshold approach	Yes	Drain crosses 50% on the fourth transfer; account promoted to CRITICAL.
5	Intervention	Yes	Automatic HOLD; customer notification on Alerts & Cases; demo-time label in the Policy Gate.
6–8	Investigation, Gemini explains, network expands	No	Depend on modules 5–7.
9	Analyst acts	API only	Fraud confirmation exists as an API; needs UI buttons.
10	Report generated	No	Depends on module 8.

5. Technical and compliance checks

Requirement	Status	Evidence
Policy gate never calls a model; model never decides (T§12, M§8.4)	Met	Lint boundary rejects any AI or intelligence import from the eight control-path modules; the decision path has no model gateway.
Data model (T§6): 11 collections, money in cents, immutable contributions, append-only interventions, lifecycle counted once	Met	Collections and indexes created automatically at start-up; verified on Atlas by the integration suite.
Intervention state machine and audit trail (T§4)	Mostly	All transitions recorded with actor; cascade hold of subsequent outflows missing.
Frontend (T§5): standalone, OnPush, NgRx / Signals / RxJS split, bufferTime(250), Web Worker, tape capped at 200	Met	Angular 22 (ADR-001). Tape is capped, not virtualised. The performance panel behind a key combination is not built.
Streaming (T§6): Pub/Sub push with OIDC, ordering key, change streams to WebSocket, snapshot on reconnect, 60-minute timeout	Met in code	Pub/Sub path tested with local push; not yet exercised on Google Cloud.
Demo time labelled when compressed (T§4, T§12 #5)	Met	"demo time ×3600" appears in the Policy Gate only when the recorded hold is shorter than an hour.
Headless DEMO_TAKEOVER_01 assertion (C§7)	Met	14 assertions: score 95, contributions sum to the score, threshold met on the fourth transfer, HELD, six codes and weights, counted once, idempotent retry, auto-release.
Latency budgets (T§10): control path under 500 ms	Partly measured	WebSocket frame latency is measured live in System Status; end-to-end control-path latency is not yet instrumented.

Deployed on Cloud Run with a public URL (C§2)	**Not yet**	Dockerfiles, Cloud Run service definitions and `cloudbuild.yaml` exist; nothing has been deployed.
Meaningful use of Google AI (C§2)	**Not yet**	See module 6.
Security posture (C§8, T§10)	**Met**	No secrets in the repository (checked); `.env` ignored; CORS limited to the UI origin; synthetic data stated in the README's first paragraph; analyst identity recorded on fraud confirmation.
English-only submission material; fixed COSMIC / multi-tenant wording (C§2, M§3, M§13)	**Met so far**	All code, docs and UI text in English; no COSMIC or real-bank claims anywhere in the UI.

6. What the application does — desktop

6.1 Operations Control Tower

The landing surface and the global situational-awareness screen. It fits one desktop viewport from a 1080p monitor down to a laptop browser window, has no search, and is driven entirely by live data from the WebSocket stream.



Control Tower at 1536×864 after DEMO_TAKEOVER_01. Top bar: menu toggle, brand, live status, Singapore clock, notifications, analyst menu. Headline KPIs with trends; the Live Global Network globe (slow rotation, a route per transfer coloured by outcome, an intervention ring on the held destination); Live Transaction Stream; Risk Analysis (95, with its largest rule contributions); System Status; Recent Alerts; Threat Landscape; Policy Gate (HOLD under SG-RD-50, 65.4% against the 50% line, hold released by the system, demo-time label).



1920×1080 monitor.



1366×640 laptop browser window: compact tier, still one screen.

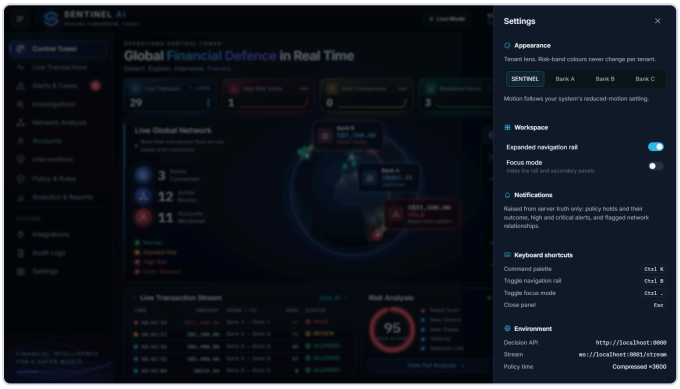
6.2 Notifications, analyst menu, settings and the navigation rail



Notifications are raised from server truth only — hold placed and released, critical alert, rapid-drain pattern, flagged network relationship. Clicking one marks it read; "Mark all read" clears the badge. Nothing navigates away.



Analyst menu: Settings and Focus mode. Settings is reached here and from the rail; the top bar has no separate gear icon.

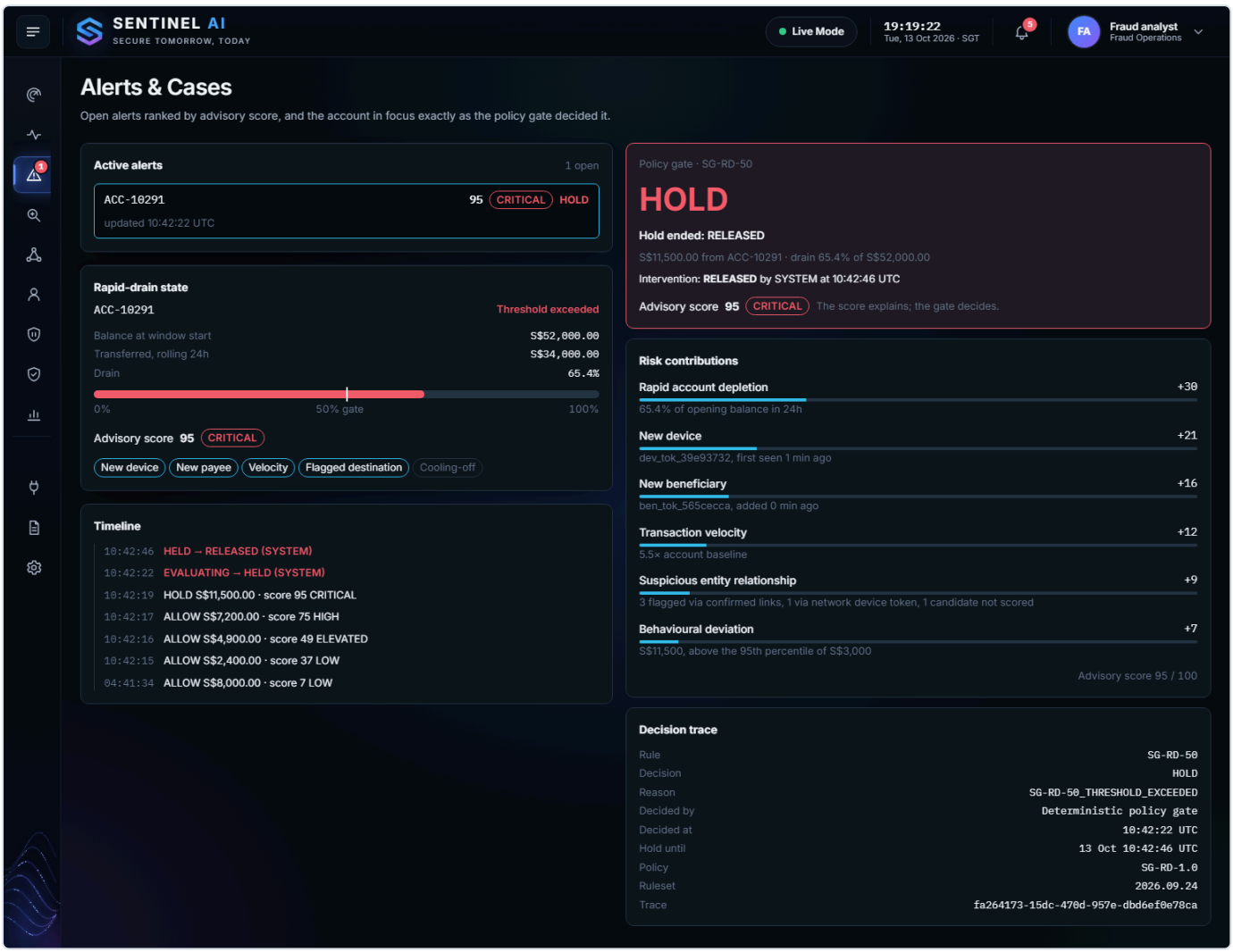


Settings drawer: tenant lens (Bank A / B / C themes), workspace preferences, notification policy, keyboard shortcuts, endpoints and system status.



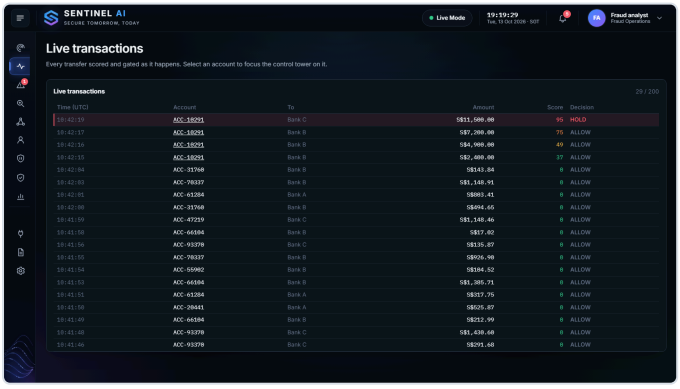
Navigation rail collapsed from the single top-left toggle; the alert badge stays visible on the Alerts & Cases icon, with tooltips on hover.

6.3 Alerts & Cases — the account in focus

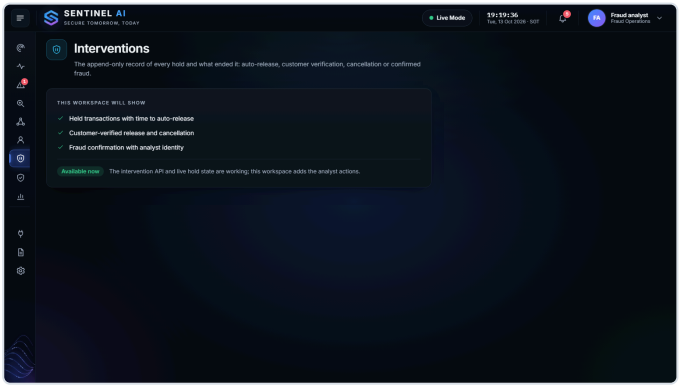


Alert queue (one open alert per account, highest risk first); rapid-drain state (balance at window start S$52,000, S$34,000 moved, 65.4% against the 50% line, the five signals); the policy gate's decision with intervention state; every rule contribution with the value that fired it, summing to 95; the decision trace (rule, reason, decided by the deterministic policy gate, policy and ruleset versions, trace ID); and the account timeline from EVALUATING → HELD → RELEASED.

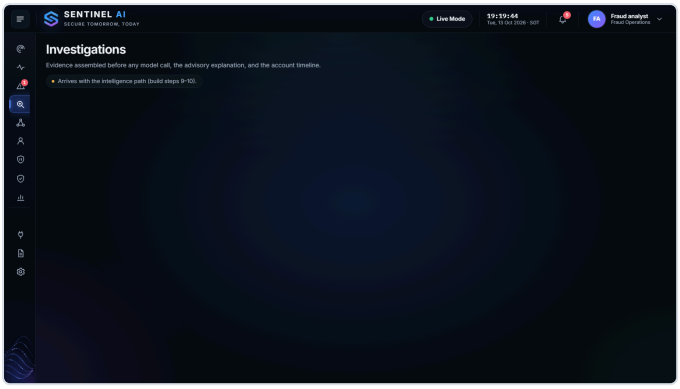
6.4 Other workspaces



Live Transactions: the full scored tape, up to 200 rows, held rows marked.



Workspaces not yet built state plainly what they will show and what already backs them.



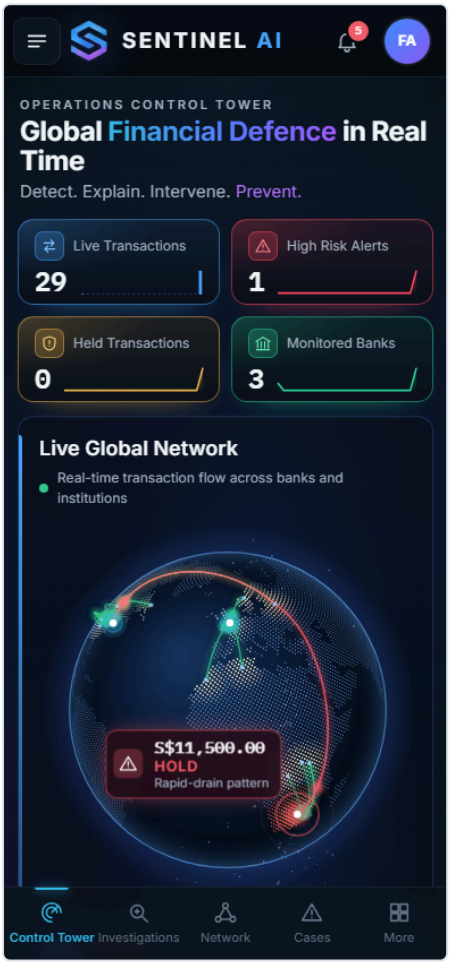
Investigations: shell awaiting the intelligence path.



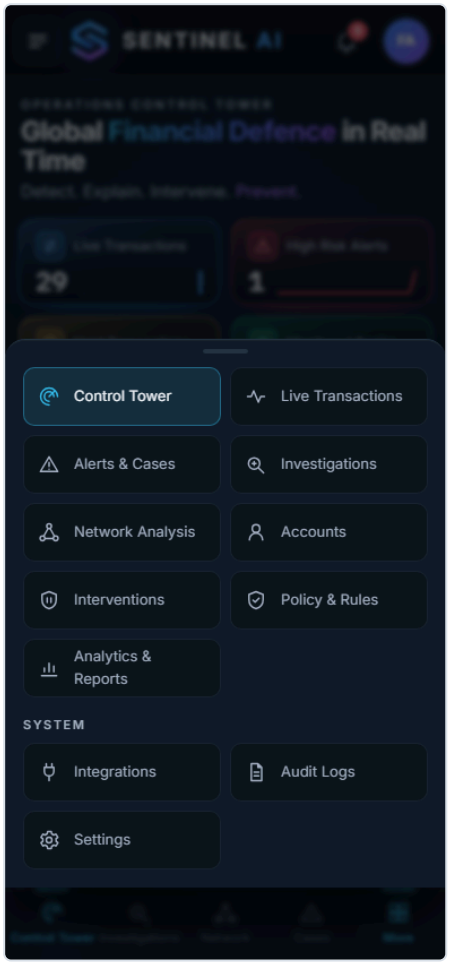
Network Analysis: shell; the graph already feeds ENTITY_LINK scoring.

7. What the application does — tablet and mobile

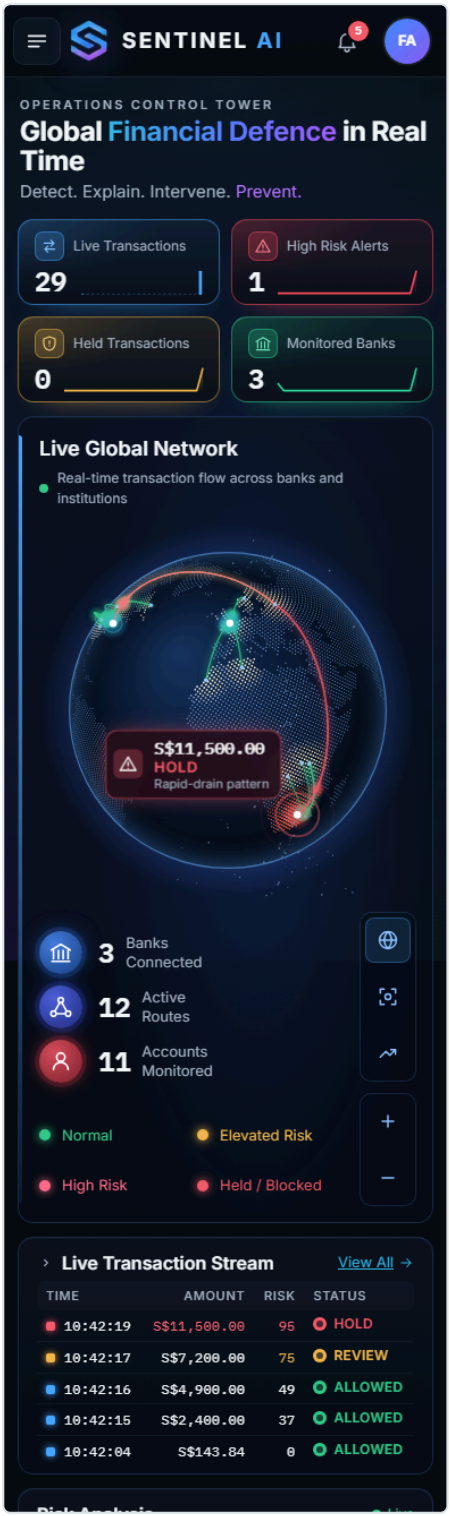
On a tablet the rail starts collapsed and panels stack. On a phone the layout is not a shrunken desktop: a compact top bar, a bottom navigation bar (Control Tower, Investigations, Network, Cases, More), a full navigation sheet with large touch targets, and a globe that keeps only the held-transfer callout so labels never collide. No page scrolls horizontally.



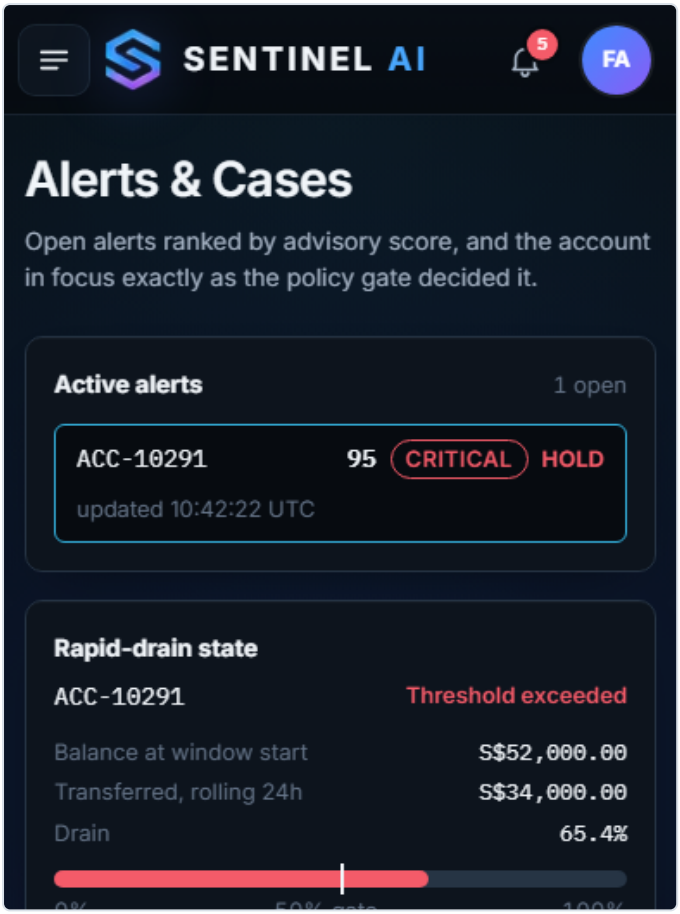
Phone, first screen (390×844).



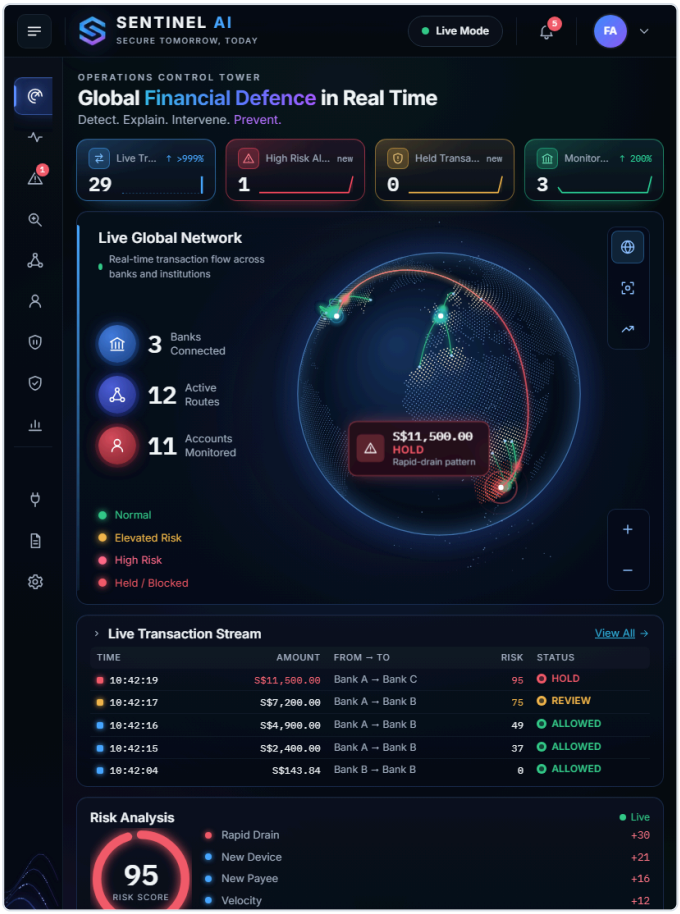
"More" opens every destination as a sheet.



Whole Control Tower page on a phone (the fixed bottom bar is hidden in this full-page capture).



Alerts & Cases on a phone.



Tablet (820×1180).

8. Phase 2 — what is pending, in the order it should be built

The documents' own rule applies: finish modules in order, and a module that renders but does not respond is worse than no module (M§11). With the freeze on 15 October and submission on 18 October, the order below puts the competition entry requirements first.

#	Work	Why now
1	Deploy the four services to Cloud Run from Cloud Build; public UI URL; Atlas access for Cloud Run	Entry requirement (C§2); "deployed, publicly reachable URL" is the first submission checklist item (M§12).
2	Gemini explanation and case summary on Vertex AI: evidence bundle, structured output, schema and `sourceField` validation, cached explanation for the demo account	Entry requirement "meaningful use of Google AI"; 40% Technical Merit & GenAI weight (M§4, T§7).
3	Investigation workspace (module 5) showing the bundle and the case summary	Demo scenes 6–7.
4	Intervention actions in the UI: release after customer verification, kill switch, confirm fraud; hold of subsequent outflows	Completes module 4 and the published control; demo scene 9.
5	Entity graph screen: clickable nodes, edge provenance, trace funds with the observed boundary	Module 7; demo scene 8; the most memorable screen (M§5.5).
6	Case report: one click, editable, sign-off, export	Module 8; demo scene 10.
7	Gemini-outage toggle and in-app performance panel	Proves intelligence independence and real-time claims in Q&A (M§13, T§5).
8	Submission package: README run-through from a clean clone, architecture diagram, proposal PDF, 3-minute video, screenshots of all six surfaces, tag `submission-v1`	M§12 checklist; C§2 deliverables.

> Items 1 and 2 are the minimum for an eligible entry. If time runs short, follow M§9: drop the entity-graph screen before anything else, and keep the seeded demo and a recorded fallback video ready.

9. Phase 1 changes made today

- Removed the "Synthetic data" and "Demo environment" wording from the top bar and analyst menu; the live indicator is a status, not a menu.
- Removed the duplicate Settings gear from the top bar (Settings remains in the analyst menu and the rail).
- Clicking a notification marks it read and nothing else.

- Compact KPI cards; four height tiers so the whole Control Tower fits from 1080p down to a 1366×640 laptop window; smaller globe; callouts never overlap.
- Collapsed-rail alert badge fully visible; Settings always visible in the rail.
- "demo time ×3600" shown in the Policy Gate only when the hold is compressed, as T§4 requires.
- Moved the runtime to Node 24 LTS; added `npm run doctor` and `npm run demo` (committed as `26ebe5d`).

10. Test results

Suite	Result
Domain · Rules · Scenarios	7 · 62 · 13 passed
Core (unit, contracts, API, DEMO_TAKEOVER_01 scenario)	73 passed
Stream · Simulator · UI	7 · 6 · 51 passed
Atlas integration (repository contracts, API smoke, change streams)	23 + 2 passed, throwaway database dropped afterwards
Typecheck, lint with architecture boundaries, production builds of all apps	Clean

All data shown is synthetic. Bank A, Bank B and Bank C are fictional participating institutions; their placement on the globe is illustrative.